

Chhattisgarh Swami Vivekananda Technical University, Bhilai (C.G.)

Program / Semester: B.Tech (IV)	Branch: Computer Science & Engineering
Subject: Computer System Architecture	Course Code: B022412(022)
Total / Minimum-Pass Marks (End Semester Exam): 100 / 35	L: 2 T: 1 P: 0 Credits: 3
Class Tests & Assignments to be conducted: 2 each	Duration (End Semester Exam): 03 Hours

UNIT- I Basic Structure Of Computers: Functional units, Basic operational concepts, Bus structures Addressing modes, subroutines: parameter passing, Instruction formats, BASIC PROCESSING UNIT: bus architecture, Instruction Cycle, sequencing of control signals, Hardwired control, Micro programmed Control, microinstruction format.

UNIT-II Arithmetic: Number representations and their operations, Design of Fast Adders, Signed multiplication, Booth's Algorithm, bit-pair recoding, Integer Division, Floating point numbers and operations, guard bits and rounding.

UNIT- III The Memory System: Various technologies used in memory design, Memory Hierarchy: Main Memory, Auxiliary Memory, Associative Memory, Cache memory, Virtual Memory. Memory Management Hardware, Multi-module memories and Interleaving,

UNIT-IV Input/ Output Organization: Peripheral Devices, I/O interfaces I/O-mapped I/O and memory-mapped I/O, interrupts and interrupt handling mechanisms, vectored interrupts, synchronous vs. asynchronous data transfer, Direct Memory Access

UNIT-V Pipeline and Vector Processing: Parallel Processing, Pipelining, Arithmetic pipeline, Instruction Pipeline, RISC Pipeline, Vector Processing, Array Processors.

Text Books:

1. Computer Organization, V.C.Hamacher,Z.G.Vranesic and S.G.Zaky, McGraw Hill,5th Edition,2002.
2. Structured Computer Organization, A.S.Tanenbaum, 4th Edition, Pearson Education India
3. Computer System Architecture, M. Marris Mano, PHI

Reference Books:

1. Computer System Architecture & Organisation, Dr. Usha, Wiley India
2. Computer Architecture & Organization, 3rd Edition, J.P. Hayes, McGraw-Hill.
3. Computer Organization & Architecture, W. Stallings, Pearson Education India

Course Outcomes [The students should be able to]:

1. Identify the basic hardware components of a computer system.
2. Familiarize themselves with binary and hexadecimal number systems including computer arithmetic.
3. Familiarize themselves with functional units of the processor such as the register file and arithmetic logical unit.
4. Understand basics functionality of systems: parallel, pipelined, superscalar and RISC/CISC architectures.
5. Represent system design in appropriate formats; addressing modes, an instruction sets as per the system configuration requirements.



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**B. Tech. (Fourth Semester) Examination,
Nov-Dec 2024
(New Scheme)**

(Computer Science and Engineering Branch)

COMPUTER SYSTEM ARCHITECTURE

Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks : 35

Note : Attempt all questions. Part(a) of each unit is compulsory. Attempt any two parts from (b), (c) and (d). Part(a)-4 marks each, Part (b),(c), (d)-8 marks each.

UNIT-I

1. (a) Explain the basic instruction format.
- (b) What are the addressing modes? Explain with suitable example.

- (c) Explain the bus architecture with suitable diagram.
- (d) Write short notes on :
 - (i) Hardwired control
 - (ii) Micro programmed control

UNIT-II

2. (a) Define arithmetic overflow.
- (b) Explain Booth multiplication along with 2's complement number using flow chart and example.
- (c) Explain the addition and subtraction of two fixed point signed magnitude data with flowchart.
- (d) Explain integer division with example.

UNIT-III

3. (a) Describe the memory types available in computer system.
- (b) A CPU has a 12 bit address for memory addressing :
 - (i) What is the memory addressability of computer?
 - (ii) If the memory has a total capacity of 16KB, What is the word length of the memory?

- (c) Explain the working flow of Virtual memory with suitable example.
- (d) What is the concept of cache memory? How it increase the speed of computer system?

UNIT-IV

4. (a) What are the peripherals devices?
- (b) Differentiate between synchronous and a synchronous data transfer.
- (c) What is DMA controller? Who control the buses on at the time of data transfer in different units of computer system?
- (d) Differentiate between I/O mapped I/O and memory mapped I/O with suitable example.

UNIT-V

5. (a) How to improve the throughput in pipelining? Write down the mechanisms to remove the collision in pipelining.
- (b) Describe the Instruction Design Pipelining and also explain the Branch handling Techniques.



- (c) What are the various pipelining hazards caused by resource usage conflicts in various instructions? Also discuss the methods to remove them.
- (d) Explain the arithmetic pipeline architecture with suitable example.



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Unit-I

1. (a) Describe various functional units of a computer.
- (b) Explain various instruction format with examples.



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- (c) With the help of block diagram explain the working of micro programmed control unit.
- (d) An instruction is stored at location 300 with its address field at location 301. The address field has the value 400. A processor register R1 contains the number 200. Evaluate the effective address if the addressing mode of the instruction is (i) direct (ii) immediate, (iii) relative, (iv) register indirect (v) index with R1 as the index register.

Unit-II

2. (a) Explain bit-pair recoding with example.
- (b) With the help of suitable logic diagram, describe how to design fast adders.
- (c) Explain Booth's Algorithm for multiplication of signed 2's complement numbers with the help of flow chart.
- (d) Explain various floating point arithmetic operations with example.



Unit-III

3. (a) Explain primary and auxiliary memory.
- (b) Draw and explain memory hierarchy.
- (c) Explain direct and set associative mapping used in cache organization.
- (d) Explain the organization of virtual memory and address translation.

Unit-IV

4. (a) Differentiate between synchronous and asynchronous data transfer.
- (b) Differentiate between IO mapped IO and Memory mapped IO.
- (c) Explain different types of interrupts and interrupt handling mechanism in detail.
- (d) Explain the working of Direct Memory Access controller with the help of block diagram.

Unit-V



5. (a) Specify a pipeline configuration to carry out arithmetic operation $(A_i * B_i) + C_i$.
- (b) Describe Flynn's classification of parallel processing.
- (c) Consider the execution of the program 15000 instructions a linear pipeline with a clock rate of 25 MHz. Assume that the instruction pipeline has 5 stages and that one instruction is issued per clock cycle. Calculate Speed up factor, Efficiency and Throughput.
- (d) Describe the working of vector processor and array processor with example.



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Roll No. :

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**B. Tech (Fourth Semester) Examination,
April-May 2021**

(Computer Science and Engg. Branch)

COMPUTER SYSTEM ARCHITECTURE

Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks : 35

Note : Attempt all question. Part (a) of all questions is compulsory. Attempt any two parts from (b), (c) and (d). The figure in the right-hand margin indicate marks.

Unit-I

1. (a) Who controls the buses in DMA data transfer and how? 4
- (b) What do you understand by a subroutine? Discuss about parameter passing in subroutines. 8

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- (c) Write the differences between Hardwired and Micro programmed control unit. Draw the block diagram of both. 8
- (d) An instruction is stored at location 300 with its address field at location 301. The address field has the value 400, a processor register $R1$ contain the number 200. Evaluate effective address if the addressing mode of the instruction are 8
- direct
 - immediate
 - relative
 - register indirect

Unit-II

2. (a) In a computer with 48-bit words, one bit is reserved for the sign. What will be the range of fixed-point integer number? 4
- (b) Describe a technique used to make the process of addition and subtraction by 2's complement number faster. 8
- (c) Explain Booth multiplication alongwith for 2's complement number using flow chart and example. 8

[3]

- (d) Evaluate the following by using Restoring Division: Divisor is 11 and Divided is 1000. 8

Unit-III

3. (a) Define the principle of locality of references. Which parameter is used to evaluate the performance of cache memory? 4
- (b) The access time of cache memory is 100 ns and that of main memory 1000 ns. It is estimated that 80% of the memory request for read and remaining for write. The hit ratio for read access only is 0.9. A write through procedure is used. 8
- What is the average access time of system considering only memory read?
 - What is the average access time of system for both read and write required?
- (c) What do you mean by virtual memory? An address space is specified by 24 bits and corresponding memory space by 16 bits. 8
- How many word are there in the address space?
 - How many words are there in the memory space?
 - If a page consists of 2K words, how many pages and blocks are there in the system?
- (d) Explain the working of associative memory with

[4]

block diagram and derive the expression for match logic. 8

Unit-IV

4. (a) How Daisy Chaining priority interrupt works? 4
- (b) Explain the direct memory access scheme alongwith its advantages for data transfer between external devices and the main memory. 8
- (c) Explain in detail about the structure of a magnetic disk system. Also mention how we can find its capacity. 8
- (d) What do you understand by computer peripherals? Explain with proper explanation any two computer peripherals. 8

Unit-V

5. (a) Specify a pipeline configuration to carry out arithmetic operation $(A_i + B_i)(C_i + D_i)$ 4
- (b) Consider the execution of the program 15000 instruction a linear pipeline processor with a clock rate of 25 MHz. Assume that the instruction pipeline has 5 stages and that one instruction is issued per

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- clock cycle. Calculate 8
- (i) speed up factor
 - (ii) efficiency
 - (iii) throughput.
- (c) What do you understand by parallel processing? Describe Flynn's classification of parallel processing. 8
 - (d) What is the use of pipelining? Prove that an M-stage linear pipeline can be at most M times faster than that of non-pipelined serial processor. 8

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Roll No. :

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**B. Tech (Fourth Semester) Examination,
Nov.-Dec. 2021**

(CSE Branch) AICTE

COMPUTER SYSTEM ARCHITECTURE

Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks : 35

*Note : Attempt all questions. Attempt any two parts
from (a), (b) and (c) of each unit. Each part
carries 10 marks.*

Unit-I

1. (a) What is Addressing Mode? Explain the types of Addressing modes in detail.

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- (b) Explain Hardwired control and Micro programmed control with neat diagram in detail. Also differentiate between Hardwired control and Micro programmed control.
- (c) Explain microinstruction format with its field in detail.

Unit-II

- 2. (a) Explain Booth's algorithm in detail. Multiply the two numbers 23 and -9 by using the Booth's multiplication algorithm.
- (b) Explain Storing and Non-Restoring method of integer division in detail. Perform Division Restoring Algorithm : Dividend = 11 Divisor = 3.
- (c) Explain Number representation with its types and operations in detail.

Unit-III

- 3. (a) What are the different types of mapping used in cache organization? Explain each in detail.
- (b) Draw and explain memory hierarchy. Explain different types of memory with neat diagram.

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- (c) Explain Multi-module memories and interleaving with example.

Unit-IV

- 4. (a) Explain I/O mapped I/O and Memory mapped I/O in detail.
- (b) What is Interrupt? Explain different types of interrupt and interrupt handling mechanism in detail.
- (c) What is DMA? Explain the working of DMA with advantages and disadvantages.

Unit-V

- 5. (a) What is Parallel Processing? Explain the concept of Vector processing in detail.
- (b) Explain the concept of Pipelining. Explain various types of pipelining in detail.
- (c) Describe Flynn's Classification of parallel processing with neat diagram and example.

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**B. Tech (Fourth Semester) Examination,
April-May 2022
(AICTE Scheme)
(CSE Branch)**

COMPUTER SYSTEM ARCHITECTURE

Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks : 35

Note : Attempt all questions. Part (a) in each unit is compulsory. Attempt any two from (b), (c) and (d). Part (a) is of 4 marks and part (b), (c) and (d) are of 8 marks each.

Unit-I

1. (a) What do you mean by micro program?
(b) Explain various types of computer instruction formats.

[2]

A two-word instruction is stored in memory at an address designated by symbol W. The address field of the instruction (stored at W+1) is designated by symbol Y. The operand used during the execution of the instruction is stored at an address symbolized by Z. An index register contains the value X. State how Z is calculated from the other addresses if the addressing mode of the instruction is :

- (i) Direct
 - (ii) Indirect
 - (iii) Relative
 - (iv) Indexed
- (c) Differentiate between hardwired control unit and micro-programmed control unit.
- (d) Explain the working of a typical hardwired control unit with a neat diagram?

Unit-II

2. (a) What is divide overflow? Give suitable example.
- (b) Show the contents of registers E, A, Q and SC

[3]

during the process of division of 10100011 by 1011 using restoring method.

- (c) Show step-by-step multiplication process using Booth Algorithm. Assume 5 bit registers that hold signed numbers are $(+15) \times (-13)$.
- (d) Explain Bit-pair recoding with an example.

Unit-III

3. (a) Write the advantages of memory interleaving?
- (b) A two-way set associative cache memory uses blocks of four words. The cache can accommodate a table of 2048 words from main memory. The main memory size is $128 \text{ K} \times 32$.
- (i) Formulate all pertinent information required to construct the cache memory.
 - (ii) What is the size of cache memory?
- (c) The access time of cache memory is 100 ns and that of main memory 1000 ns. It is estimated that 80% of the memory request for read and remaining 20% for write. The hit ratio for read access only 0.9. A write through procedure is used.

[4]

- (i) What is the average access time of the system considering only memory read cycle?
- (ii) What is the average access time of the system for both read and write required?
- (iii) What is the hit ratio taking into consideration the write cycles?
- (d) Explain the working of an associative memory with neat diagram.

Unit-IV

- 4. (a) What are priority interrupts.
- (b) Explain the working of Direct Memory Access with a neat diagram.
- (c) Differentiate between synchronous and asynchronous data transfer.
- (d) Explain programmed I/O with example.

Unit-V

- 5. (a) Define parallel processing?
- (b) Explain Flynn's classification of parallel architecture.

[5]

- (c) What do you understand by pipelining? Differentiate between instruction pipeline and arithmetic pipeline.
- (d) A nonpipeline system takes 50 ns to process a task. The same task can be processed in a six-segment pipeline with a clock cycle of 10 ns. Determine the speed up ratio of the pipeline for 100 tasks. What is the maximum speed up that can be achieved?



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(New Scheme)

(CSE Branch)

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Maximum Marks : 100

Minimum Pass Marks : 35

Note : Attempt all questions. Part (a) in each unit is compulsory. Attempt any two from (b), (c) and (d). Part (a) is of 4 marks and part (b), (c) and (d) are of 8 marks each.

Unit-I

1. (a) What do you understand by addressing modes?

- (b) Write the differences between hardwired and micro programmed control unit. Draw the block diagram of both.
- (c) Explain the representation, selection and application of instruction formats
- (d) Explain accumulator based CPU with proper diagram.

Unit-II

- 2. (a) Define terms overflow and underflow.
- (b) Write short notes on :
 - (i) Fast address
 - (ii) Bit pair recording of multipliers
- (c) Explain the floating point operations with example. Define guard bits
- (d) Describe Booth Multiplication algorithm with example

Unit-III

- 3. (a) Define Cache Memory.
- (b) Explain the associative memory organization and derive the expression for match logic.
- (c) Explain virtual memory in details.
- (d) How many 128 byte RAM chips are required to provide a memory of 2048 bytes show details connection clearly indicating address, data and decoder?

Unit-IV

- 4. (a) Define peripheral devices.
- (b) What is direct memory access technique? Explain the rate of DMA controller with diagram.
- (c) Write down the differences between I/O mapped I/O and memory mapped I/O
- (d) Explain the working of daisy chaining priority interrupt with diagram.

Unit-V

5. (a) Define pipelining.
- (b) Draw and explain flow chart and timing diagram for the four segment instruction pipeline.
- (c) Define parallel processing. Describe Flynn's classification of parallel processing.
- (d) Write short notes on :
- (i) Vector processing
 - (ii) Array processing.

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**B. Tech. (Fourth Semester) Examination,
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(Computer Science and Engineering Branch)

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Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks : 35

Note : Attempt all questions. Part(a) of each unit is compulsory. Attempt any two parts from (b), (c) and (d). Part(a)-4 marks each, Part (b),(c), (d)-8 marks each.

Unit-I

1. (a) Explain register indirect mode.

4

(b) Evaluate the arithmetic statement $X=(A+B)*(C+D)$

using three address instruction and two address instruction.

8

(c) What is the format of microinstruction in computer architecture?

8

(d) Explain Instruction Cycle with diagram.

8

Unit-II

2. (a) Defined Signed number.

4

(b) Multiply (-5) and (4) using Booth's Algorithm.
(Register size = 4 bit)

8

(c) Explain Bit Pair Recoding and perform multiply $(+13)$ and (-6) using bit pair recoding multiplier method.

8

(d) Explain Restoring division and divide 17 by 3 using restoring method.

8

Unit-III

3. (a) What is Virtual Memory?

4

(b) A direct mapped cache of size 64 word, with block

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size 4 word, the size of main memory is 512 Byte.

Find :

- (i) Physical address
- (ii) Block offset 8
- (c) Write memory hierarchy with diagram. 8
- (d) The access time of a cache memory is 100 ns and that of main memory is 1000 ns. It is estimate that 80 percent of the memory request are for read and remaining 20 percent for write the hit ratio for read access is 0.9. A write through proeedure is used :
 - (i) What is the average access time of the system considering only memory read cycle?
 - (ii) What is the average access time of the system for both read and write cycles?
 - (iii) What is the hit ratio taking into consideration the write cycle? 8

Unit-IV

4. (a) Explain Peripheral devices. 4

- (b) Difference between synchronous and asynchronous data transmission. 8
- (c) Draw the block diagram of DMA controller and explain. 8
- (d) What are Interrupt? Explain various types of interrupt. 8

Unit-V

- 5. (a) Explain RISC Pipeline. 4
- (b) Describe Pipeline Processing. 8
- (c) Explain Processor level parallelism. 8
- (d) Explain arithmetic pipeline with suitable example. 8